

Bounds on the Dynamic Causal Effects of Monetary Policy

An analysis using the
Federal Reserve's Textual Repository

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Abstract

Monetary policy – the Federal Reserve's actions to achieve maximum employment and stable prices – is a fundamental driver of the global economy. Understanding and quantifying its effects has been and still is a key area in macroeconometrics. Inspired by recent literature on semi and non-parametric approaches to the issue of estimating dynamic causal effects of monetary policy, this paper uses a novel data constructed from the Federal Reserve's textual repository to construct identifiable and estimable bounds on this effect under very general conditions. These bounds are informative, capture non-linear causal effects, and are in line with both the existing empirical research and established macroeconomic theory.

1 Introduction and Motivation

Many of the most important questions in empirical macroeconomics rely on dynamic causal effects, and perhaps the most important of these is that of understanding and quantifying the effects of monetary policy. Monetary policy, the Federal Reserve's actions to achieve maximum employment and stable inflation, is a fundamental driver of the United States' economy and its financial system – and, in turn, the global economy. The issue around correctly estimating the dynamic causal effect of monetary policy intervention over time has been – and still is – an active area of research in both theoretical and applied macroeconomics.

Conceptualizing and estimating the effects of monetary policy poses, of course, many challenges. Dynamic feedback between the policy interventions and the observed outcomes or covariates makes it extremely difficult to disentangle causes from effects (Rambachan and Shephard, 2019). Explicit assumptions about the timing and correlation structure of such responses to interventions are often times required, and the availability of data at monthly frequencies for at most a few decades makes the task of obtaining accurate and consistent estimates daunting. Due to these challenges, academia has often turned to parametric linear models, such as structural VARs (Sims, 1980) or local projections (Jorda, 2005). These, while tractable and intuitive, have some severe downsides: a reliance on linear models imposes extra restrictions on underlying dynamic causal effects and the processes beneath them.

For this reason, recent literature on monetary policy causal effect estimation has transitioned to non- or semi-parametric models. Angrist, Jorda and Kuersteiner (2018)¹ and Rambachan and Shephard (2019) argue that the issues surrounding identifying and estimating the effects of the Federal Reserve's interventions are similar to those commonly present in

¹ Referred henceforth as Angrist et al. (2018)

microeconomic policy evaluation with observational data. The key problem that emerges under this scenario is that different policy choices – i.e. decreasing or increasing the federal funds rate target – are more or less likely depending on what region of the covariate space you are in – i.e. high unemployment, low inflation, etc. In other words, when estimating the effects of monetary policy intervention, one faces a selection-into-treatment problem: each time period self-selects into a particular policy choice.

There is an extensive body of literature that addresses the problem of causal effects under the selection-into-treatment scenario.² Two common assumptions are usually necessary for identification and estimation under this paradigm. The first, *selection on observables* (or unconfoundedness), demands that, conditional on the covariates observed by the econometrician, any other variable that might impact the policy choice be independent of the future counterfactual evolution of the outcome variables. The second, *overlap*, requires that the probability of any given policy choice being selected is strictly positive at every element of the covariate space.

Recent studies (e.g. Angrist et al. 2018) on semi-parametric estimates of the effects of monetary policy have worked under a potential outcomes framework where they take both of these assumptions as true.³ However, this paper argues that even if one is willing to work under the selection on observables paradigm, overlap is still very likely to fail in the context of monetary policy analysis. Intuitively, the reason is that certain subsets of the covariate space preclude certain policy choices.

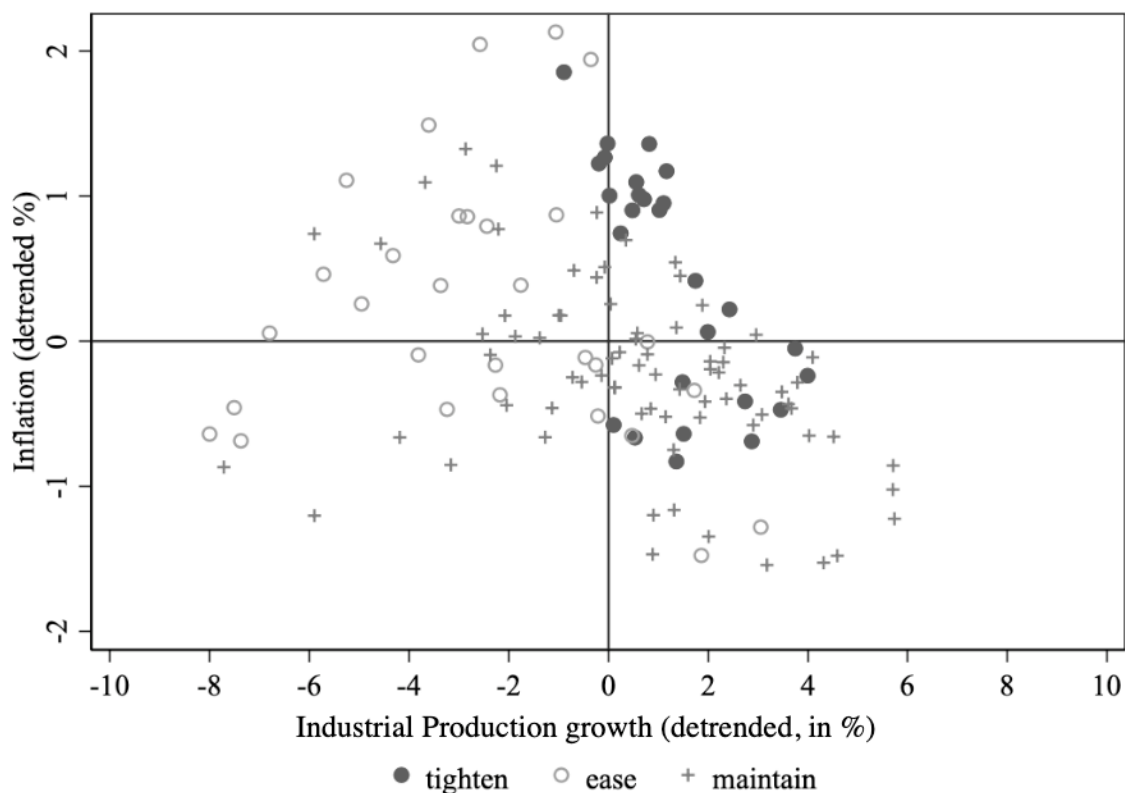
To illustrate this, assume that at each point in time, the Federal Open Market Committee (FOMC) has only three policy choices available: tighten, ease, or maintain the target funds rate.

² See Imbens and Rubin, 2015, for an excellent comprehensive survey.

³ Obviously, by justifying why that is the case both implicitly or explicitly.

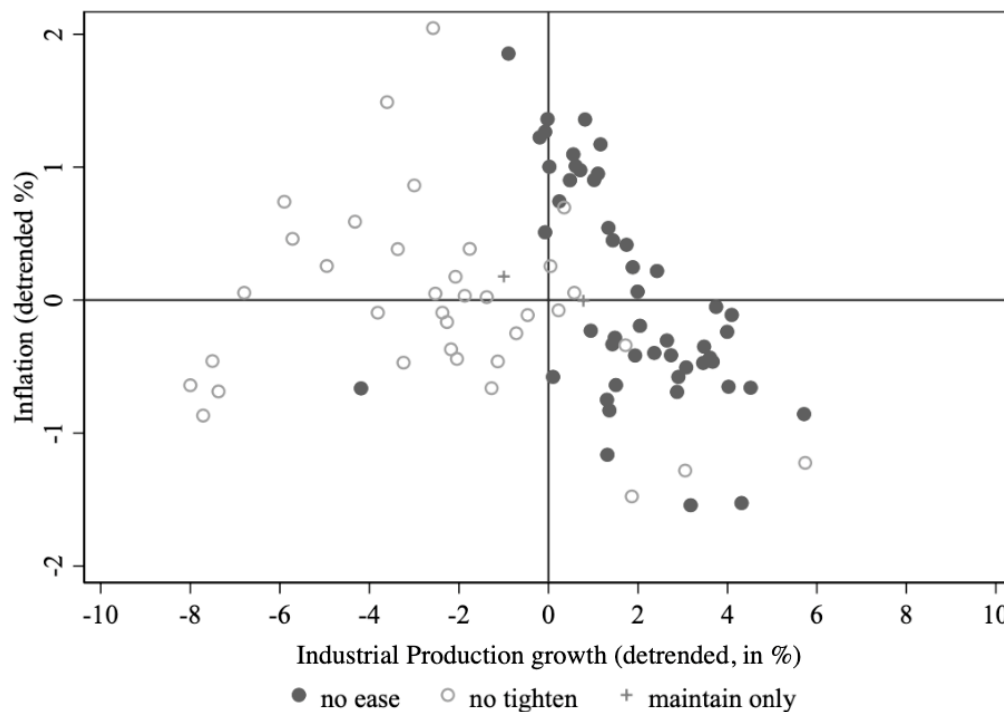
Also, suppose that the information set faced by the econometrician contains only one lag of inflation and one lag of industrial production (a proxy for economic output). As Taylor Rule macroeconomic theory suggests, an increase in the federal funds rate would slow down the economy, reducing inflationary pressure and economic output. Thus, since the goal of monetary policy is to promote high economic activity (and employment) and stable prices, high values of inflation and industrial production are unlikely to be met with a decrease in the target rate. Figure 1 illustrates that point: historically, the FOMC has not tightened (eased) its stance when both economic output growth and inflation are below (above) their long-term trends.

Figure 1: Policy Choices in Different Inflation and IP Environments



To make matters more formal and rigorous, the overlap assumption refers not to the actual policy *choices* but rather to policy *alternatives*,⁴ where latter term means the set of policy actions considered by the FOMC at any given meeting. Through the use of the Federal Reserve’s extensive textual dataset, which is described in detail later, one is able to extract information regarding the FOMC’s so-called choice set, creating a time series of what is denominated in this paper as “policy alternatives” or “policy menus”. I focus on most of the Greenspan period, spanning from June 1989 to January 2006, since during this time the federal funds rate was explicitly discussed as the main tool in setting monetary policy. Figure 2 below provides a similar representation as that of Figure 1, but now using the time-series of policy alternatives instead. Indeed, certain subsets of the covariate set (here, inflation and output) do preclude certain policy choices.

Figure 2: Policy Menus in Different Inflation and IP Environments



⁴ Assuming D_t and W_t are the policy variable and covariates, respectively, overlap refers to $P(D_t = d | W_t)$ not the realizations of D_t themselves.

This paper, through the use of this novel time-series of policy alternatives constructed from textual data, extends and generalizes the semi-parametric approach introduced by Angrist et al. (2018) by allowing the overlap assumption to be violated in a very specific fashion.⁵ In doing so, I construct identifiable and estimable bounds on the causal effects of monetary policy intervention while both assuming no restrictions on the structure of the feedback process between variables and allowing for a key identification assumption to be violated.

This paper is organized as follows. Section 2 presents a detailed description of how the time-series of policy alternatives was constructed from the textual data. Section 3 lays out the main theoretical framework for our model as well as the derivation of the bounds on the causal effects. Section 4 outlines the general estimation procedure used, discusses the estimation of the propensity score as well as its results, and explains the challenges surrounding inference in the context of our estimator. Section 5 presents and discusses the estimated bounds on the causal effect of monetary policy on three key macroeconomic aggregate variables: inflation, economic output, and unemployment. Section 6 concludes the paper and presents an overview of its contributions.

2 Data Collection: Textual Analysis

One of the innovations of this paper is the use of textual data from the Federal Reserve's repository as a means of constructing key variables, building on recent work led by Prof. Montiel Olea.⁶ This section outlines how this extensive body of data was extracted.

⁵ This will be explained further in Section 3.

⁶ Full data and code repository can be found at: <https://github.com/jm4474/FOMCTextAnalysis>

2.1 Monetary Policy decisions in the U.S.

The Federal Open Market Committee (FOMC) is the body of the Federal Reserve System in charge of setting monetary policy. Members and staff of the FOMC meet regularly in pre-scheduled meetings to discuss a number of issues surrounding the domestic and international economy that can be seen as relevant for policy. The purpose of this meeting is, by means of a discussion, agree on the stance of American monetary policy.

In preparation for these regularly scheduled meetings, typically within every six weeks, the Federal Reserve's staff will produce a variety of documents – a process that is highly structured. At the expense of rigor and at risk of oversimplifying, this process as well as the documents that are produced within it can be summarized as follows:

- (i) Two weeks before the meeting, the *Beigebook* is released: this summarizes empirical and anecdotal information on the current economic scenario in each of the twelve Federal Reserve districts.
- (ii) One week before the meeting, the *Greenbook* is distributed to FOMC members: this provides the Federal Reserve's outlook for the domestic and global economy
- (iii) Typically on the Friday before the meeting, the staff at the Board of Governors circulates the *Bluebook*: this lists and provides context on the FOMC's policy alternatives for the upcoming meeting.
- (iv) At last, the FOMC secretariat and Chairman produce an *agenda* for the meeting: as put by the Federal Reserve's website, "the agenda includes the standard topics covered at each such meeting (for example, reports on open market operations, the economic situation, and monetary policy alternatives), as well as any special items".

This is a necessarily course and brief overview, but necessary for the discussion of the data extraction process in this paper.⁷ It is also worth noting that the format and manner in which the FOMC and the Federal Reserve has released these documents over time. Documents have been introduced, merged, expanded or disregarded continuously from the beginning of the institution's text corpus in 1936. Figure A.1.1 in the appendix provides a visual summary of this evolution. After all these documents are released to the appropriate members of the Federal Reserve, the meeting occurs, and the FOMC finally communicates its stance on monetary policy to the public.⁸

2.2 Selection and Extraction of the Data

Over the years, the Federal Reserve releases the documents listed above and makes them readily available for download in their website.⁹ After downloading and extracting the selected documents, in a process of converting them from Hypertext Markup Language (HTML) to raw text, a variety of commonly-available techniques in Natural Language Processing (NLP) can be used to construct our required dataset, amongst them *regular expressions*. These are a sequence of characters that define a search pattern, and are widely used to in studies like this one. A simple example of the use of a regular expression in Python could be to find all occurrences of the word “increase” or variations such as “increases” or “increased” in a sequence of words stored under the variable **text**, as follows:

⁷ For a more detailed explanation, see: <https://www.philadelphiafed.org/education/teachers/resources/day-in-life-of-fomc#02>

⁸ Before 1994, the decision was not readily announced to the public. Between May 1994 and January 1994 the Record of Policy Actions was the only timely communication, made available to the public around 30 days after the meeting. This became what we know as the *minutes*. Also, since 1994, the FOMC secretariat started to prepare for each meeting a verbatim report of discussions, known as the *transcript*, and made available to the public with a long lag of five years.

⁹ https://www.federalreserve.gov/monetarypolicy/fomc_historical_year.ht

```
re.search("increas\w+", text")
```

Without going too much into detail into the syntax and what the *metacharacters* “\w+” represent, this programming tool is crucial in the process of going from raw text to informative data points.

2.3 Construction of Policy Alternatives

As aforementioned, the Federal Reserve produces a variety of different documents at different points in time for an upcoming meeting. However, the *Bluebook* is the most crucial one for our purposes, as they devote one section to enumerate all policy alternatives available to the FOMC, as well as explaining each of their rationales. Within the context of creating a set of “policy menus”, there seems to be no better starting point.

Regular expressions, as briefly explained above, were used to extract all of these policy alternatives from the Bluebooks, due to the flexibility and automatization that it offers as a word scraping tool. The following regular expression code was used to find all sentences with any mention of a policy alternative:

```
([^\.]*)(alternatives?\s+[a-e])(\.|[^\a-z]\.|[^\a-z][^\.]+\.)
```

This procedure extracted all relevant sentences from otherwise very long documents. However, once this was done, one had to manually read the dataset containing these sentences and classify the alternatives based on an interpretation of the excerpts.¹⁰ In particular, the federal funds target rate for each policy alternative was extracted, as well as a categorization between easing, tightening and unchanged policy.

¹⁰ At times, the same policy alternative in terms of the target rate were stated in different fashions due to different rationales. Thus, one had to go back to the original text to make sense of the data and correctly classify it.

Table 1 presents a low-dimensional representation of the extracted policy menu between our selected subsample of July 1989 to January 2006. It outlines the frequencies of each set of policy alternatives over this subsample. This simplifies the menu by grouping the 25 basis points increments into these three categories (trivially), and for the purpose of this study is key.

Table 1: Summary Statistics for Target Changes (1989-2006)

Policy Options	Number of Meetings
E	3
T	7
U	1
U,E	31
U,T	43
U,T,E	49
Total	134

A more complete and sensitive study would have looked into the individual 25 basis points increments as the policy choices themselves, making the analysis of causal effects more granular and informative. However, as it can be seen in the Appendix (Table A.1.1 and A.1.2), data for the extreme case of a 50 basis points increase or decrease were scarce, threatening consistent and reliable estimation. Therefore, as will be further developed in the next section, we limited our policy choices to the three categories of easing, unchanged, and tightening.

3 Monetary Policy Evaluation: A Model

This paper is fundamentally interested in how monetary policy affects the economy through all of its available channels. Two main challenges arise when addressing this: defining what metric is better used to define the stance of monetary policy, and dealing with the dynamic nature of the problem. For this, I extend the potential outcome framework recently suggested in Angrist et al. (2018).

3.1 Defining the Treatment Variable

As mentioned in the previous two sections, monetary policy is a term in reference to the Federal Reserve's actions to promote maximum employment and maintain stable inflation levels. Prior to the recent financial crisis in 2007-08, the Federal Reserve acted mainly through open market operations – trading of securities issued by the U.S. government in the open market – in order to control a vitally important short-term (interbank) interest rate, called the federal funds rate. The Federal Reserve has perfect control over its target rate, but imperfect control over the actual federal funds rate, so it has been common in literature to assume the target rate as the policy (or treatment) variable.

This is the stance taken in this paper: I assume the Federal Reserve has a scalar and discrete policy variable, D_t , that refers to the *changes* to a given federal funds rate target. As mentioned in Section 2, even though targets are usually set to intervals of 25 basis points, I use a more condensed parameter space due to insufficient data. That is, I assume that $D_t \in \mathcal{D} = \{-1, 0, 1\}$, the elements of which correspond to decreasing, not changing, and increasing the target rate, respectively.

It is also worth noting that it has not always been the case that the Federal Reserve had an explicit target for the federal funds rate. There has been constant disagreement about the exact point in time at which this began to be the case. Romer and Romer (2004) argue that “Most obviously, for much of the period for which we have detailed forecast data (1969-1996), the Federal Reserve explicitly targeted the funds rate”. Thus, by restricting the sample to most of the Greenspan period (July 1989-January 2006), one can be confident that this was the case.

3.2 Model Set-up: Potential Outcomes Framework

A key assumption is that the Federal Reserve relies on data – both macroeconomic aggregate data as well as market conditions – when determining policy. Let the economy be denoted by $Z_t \in \mathbb{R}^k$, a finite-dimensional vector of covariates containing both lagged aggregate macroeconomic data as well as asset price information about policy actions. Also, I use the policy menus constructed as described in Section 2 to construct another covariate $X_t \in \mathbb{X} = \{\{0\}, \{1\}, \{-1\}, \{0,1\}, \{0, -1\}, \{-1,0,1\}\}$. That is, the covariate space of X_t is the span of all possible combinations in the policy menus – a set of sets. Denote the entire covariate set as $W_t = [Z_t, X_t] \in \mathbb{W}$.

Since the Federal Reserve relies on data (and obviously the policy menus extracted) to make policy decisions, let us define a flexible parametric model known as the *propensity score*:

$$P^d(W_t) = P(D_t = d | W_t)^{11}$$

Finally, let $Y_t \in \mathbb{R}$ denote any outcome variable of interest at time t – we keep it Y_t one-dimensional for the set-up as the expansion into a multidimensional outcome vector is trivial,

¹¹ I assume the propensity score does not change over time: the function $P(\cdot)$ does not explicitly depend on t .

and exposition becomes clearer without this unnecessary generalization.¹² Outcome variables that could be of interest include all macroeconomic variables affected by monetary policy: through different channels, changing the interest rate affects a variety of key aggregate and market-based variables.¹³

Assume that a policy choice $D_t = d$ induces a counterfactual distribution over the outcome variable Y_t , h periods in the future: this is modelled as a random variable $Y_{t+h}(d)$, which has some arbitrary distribution conditional on the covariates W_t . These are the *potential outcomes* for the outcome variable.

For two possible policy choices at time t , denoted as $D_t = d$ and our benchmark policy $D_t = 0$, our parameter of interest is the effect of the policy choice $D_t = d$ with respect to policy choice $D_t = 0$:

$$\theta_{t+h,d} = E[Y_{t+h}(d) - Y_{t+h}(0)]$$

Also, the realizations of the outcome variable are only observed for the specific policy choice that was actually chosen at time t :

$$Y_{t+h} = \sum_{d \in \mathcal{D}} Y_{t+h}(d) * 1\{D_t = d\}$$

As mentioned in the introductory section, two common assumptions enable identification and estimation of $\theta_{t+h,d}$ in the presence of selection-into-treatment: overlap and selection on observables. These are satisfied if and only if both conditions hold for any time period t , for all $h > 0$ and for all $w \in \mathbb{W}$:

$$Y_{t+h}(D_t) \perp D_t | W_t = w \quad (1)$$

¹² Also, I avoid the need to have a “policy regime” idiosyncratic variable ϕ as in Angrist et al. (2018) since I limit my sample to the Greenspan regime.

¹³ For our empirical analysis in Section 5, I chose to focus on three key variables, however: inflation, economic output, and the unemployment rate.

$$P^d(W_t) = P(D_t = d | W_t) > 0 \quad (2)$$

That is, the potential outcomes $Y_{t+h}(D_t)$ are independent of the policy choice D_t conditional on covariates W_t , and no element or subset of the covariate space $\mathbb{W}$ precludes a certain policy choice.

Condition (1) enables one to write the average causal effect conditional on covariates W_t in terms of observable means:

$$E[Y_{t+h}(d) - Y_{t+h}(0) | W_t] = E[Y_{t+h}|D_t = d, W_t] - E[Y_{t+h}|D_t = 0, W_t] \quad (3)$$

As in Angrist et al. (2018), average causal effects can then be estimated by using the fact that condition (1) implies that:

$$E[Y_{t+h}1\{D_t = d\} | W_t] = E[Y_{t+h}(d)|W_t] * P^d(W_t)$$

Solving the above for $E[Y_{t+h}(d)|W_t]$ and integrating over W_t , one can express the average causal effect using the Inverse Probability Weighting (IPW) estimator:

$$\begin{aligned} \theta_{t+h,d} &= E[Y_{t+h}(d) - Y_{t+h}(0)] \\ &= E\left[Y_{t+h}\left(\frac{1\{D_t = d\}}{p^d(W_t)} - \frac{1\{D_t = 0\}}{p^0(W_t)}\right)\right] \end{aligned} \quad (4)$$

As aforementioned and motivated, even if one is willing to work under the assumption of selection on observables (condition (1)), overlap is still likely to fail. This has serious implications in the above results. First, the expansion into observable conditional means in (3) is no longer feasible, as if $p^d(W_t) = 0$ for some subset $A \subset \mathbb{W}$ and for some $d \in \mathcal{D}$, one of the expectations will not be identified in the sampling process. Furthermore, under the same conditions above, the IPW estimator is no longer feasible as a fundamental zero-division error occurs.

Thus, under a violation of overlap, one needs to readjust the procedure. We approach this by violating overlap in a very specific and systematic fashion:

$$p^d(W_t) = P(D_t = d | W_t = w) = P(D_t = d | X_t = x, Z_t = z) = 0 \text{ if and only if } d \notin x \quad (5)$$

That is, the propensity score is zero when the policy choice being considered, d , is not within the policy menu $X_t = x$ at that given moment in time.

3.3 Bounds on Dynamic Causal Effects

As argued above, overlap makes the approach pioneered in Angrist et al. (2018) statistically unfeasible. Thus, I address this by developing bounds for the average causal effect of monetary policy, as opposed to point estimates. As in Manski (1990 and 2011), assume the outcome variable of interest Y_t is bounded by a finite closed set on $\mathbb{R}$.¹⁴ That is, there exists $K_1, K_2 \in \mathbb{R}$ such that for all t and all d :

$$Y_t(d) \in [K_1, K_2]$$

Note that selection on observables, condition (1), is still taken as given. Now, trivially, the following also holds true:

$$E[Y_{t+h}(d) - Y_{t+h}(0)] \in [K_1 - K_2, K_2 - K_1] \quad (6)$$

Since the outcome variables of interest are always rates of changes (as will be made clearer later), one can safely assume that $K_1 < 0$, $K_2 > 0$ ¹⁵. Thus, the above implies a bound around zero:

$$[K_1 - K_2, K_2 - K_1] \subset [-A, A] \quad , \text{ for some } A \in \mathbb{R}, A > 0$$

Expanding $\theta_{t+h,d}$ by conditioning on the policy menus X_t , we have that:

$$\theta_{t+h,d} = E[Y_{t+h}(d) - Y_{t+h}(0)] = \sum_{x \in \mathbb{X}} P(X_t = x) * E[Y_{t+h}(d) - Y_{t+h}(0) | X_t = x] \quad (7)$$

¹⁴ Since the outcome variables of interest are changes or differences, this assumption holds intuitively and empirically.

¹⁵ This assumption by no means affects our results or the derivation. It is simply due to basic intuition and for ease of exposition.

Thus, by combining condition (5) and equations (6) and (7), one can bound $\theta_{t+h,d}$ in the following interval:

$$\theta_{t+h,d} \in \left[\sum_{d \in \mathcal{D}} E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x] * P(X_t = x) + (K_1 - K_2) \sum_{d \notin \mathcal{D}} P(X_t = x), \right. \\ \left. \sum_{d \in \mathcal{D}} E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x] * P(X_t = x) + (K_2 - K_1) \sum_{d \notin \mathcal{D}} P(X_t = x) \right] \quad (8)$$

The width of the interval is thus:

$$(2K_2 - 2K_1) \sum_{d \notin \mathcal{D}} P(X_t = x) < 2K_2 - 2K_1$$

The right-hand side of the above equation is the width of the original interval (6), and therefore the bounds are necessarily informative.

These bounds, under selection on observables (condition (1)), are identified and estimable as before without the violation of overlap. The conditional expectations in (8) can be consistently estimated using the IPW in the selected subsample as we revert to the same situation as before¹⁶. Estimation of the marginal probabilities of X_t and of the initial outcome bounds K_1 and K_2 are trivial.

¹⁶ One possible concern would be if X and Z were perfectly correlated: i.e. if the information captured by the textual analysis for the policy menus captured all relevant information from Z in determining policy. Appendix A.2 shows that does not affect our estimation, and the propensity score weights displayed in Section 4.2 further support this claim.

4 Estimation and Inference

4.1 General Estimation Procedure

The key issue surrounding estimation of the bounds is implementing the IPW estimator for the conditional expectations in (8). Apart from the propensity score, which is computed first and applies to all for obvious reasons, I compute them individually in their respective subsamples in a similar fashion as in Angrist et al. (2018)¹⁷, and each can be written as weighted averages of the vector of future outcomes $\mathbf{Y}_{t+h} = (Y_{t+1}, \dots, Y_{t+h})'$ with the weights formed by:

$$\delta_{t,d} = \frac{1\{D_t = d\}}{p^d(W_t)} - \frac{1\{D_t = 0\}}{p^0(W_t)}$$

In a correctly specified model, which should resemble a randomized experiment, these weights should be mean zero and uncorrelated with the covariates W_t . Therefore, I ensure this is true by regressing $\delta_{t,d}$ on the covariates and a constant, and using the residual weights, which we denote as $\ddot{\delta}_{t,d}$, instead. The general estimation procedure can be outlined as follows:

- 1) Estimate the propensity score, $P(D_t = d | W_t)$ and compute the predicted probabilities $\widehat{p}^d(W_t)$ for all d and all t .
- 2) Construct the weights $\delta_{t,d}$ and residualize them to obtain $\ddot{\delta}_{t,d}$.
- 3) For the given policy choice at hand, $D_t = d \in \{-1, 1\}$, compute the vector of estimated responses in the appropriate subsample, with observations corresponding to those with $t \in T(d) \subset \{1, \dots, T\}$:

$$\widehat{\theta}_d = \frac{\sum_{t \in T(d)} \mathbf{Y}_{t+h} \ddot{\delta}_{t,d}}{|T(d)|}$$

¹⁷ There are two key differences in our approach. Firstly, I do not have to drop observations with very low propensity scores as in Angrist et al. (2018) due to my explicit treatment of overlap. Furthermore, I need to compute the individual responses on the relevant subsample.

Where the denominator is the size of the subsample.

- 4) Estimate the marginal probabilities $P(X_t = x)$ and the outcome bounds K_1, K_2 .
- 5) Compute the bounds as in (8).

4.2 Propensity Score Estimation

In line with the literature on multi-level propensity score estimation (Angrist et al. 2018, Angrist et al. 2011), I use an ordered probit specification to estimate the propensity score as outlined in the first step of Section 4.1. The choice of covariates was inspired by a number of factors, which are represented iteratively in the specifications in Table 2 (and Table A.3.2). Firstly, I included as the main covariates inflation (measured by personal consumption expenditures price index, PCEPI), the unemployment rate and industrial production, which commonly appear in Taylor Rule inspired empirical studies. Next, as in Angrist et al. (2018), I added changes in the prices of federal funds rate futures as a description of market-based information on what policy the FOMC will hold.¹⁸ Finally, I add dummies of the policy menus constructed from the textual data. For all three specifications, we also include a number of control variables, including a set of monthly dummies to control for seasonal variation as well as dummies for idiosyncratic events (e.g. 9/11).

¹⁸ For a more detailed explanation on the matter, read Section 3 from Angrist et al. (2018)

Table 2: Ordered Probit Specifications (Marginal Effects on Increase)

	(1)	(2)	(3)
Maintain in Menu			-0.023 (0.044)
Ease in Menu			-0.089** (0.043)
Tighten in Menu			0.139*** (0.033)
FFF		0.902 *** (0.123)	0.583*** (0.097)
Inflation, Lag 1	0.127 (0.115)	0.136 (0.092)	0.095 (0.076)
Inflation, Lag 2	0.194* (0.117)	-0.016 (0.100)	-0.026 (0.082)
Unemp. rate, Lag 1	-0.376*** (0.136)	-0.214** (0.107)	-0.166* (0.087)
Unemp. rate, Lag 2	-0.537*** (0.139)	-0.247** (0.113)	-0.151 (0.094)
IP, Lag 1	0.118*** (0.039)	0.071** (0.033)	0.031 (0.027)
IP, Lag 2	-0.111*** (0.039)	-0.065** (0.032)	-0.027 (0.027)
Log Likelihood	-127.44	-88.39	-72.53
Observations	134	134	134

Note: Table reports marginal effects of listed covariates on the probability of an increase of the target rate. */**/** corresponds to significance at the 90/95/99% confidence level.

Table 2 displays the marginal effects of the different covariates on the probability of a rate increase for all three specifications.¹⁹ Initially, both unemployment and industrial production are significant predictors of the target change at the 99% confidence level. Once a market-based covariate (FFF) is added to the model, their significance drops to the 95% level, as does the magnitude of the marginal effect. The inclusion of FFF also dramatically improves the model fit, as the likelihood change indicates. However, this is even further improved once the dummies for

¹⁹ Equivalent table for a rate decrease is displayed in the Appendix (Table A.3.2). Results are analogous.

the policy menus are added. Only the first lag of unemployment remains significant out of the fundamental variables, and the inclusion of tightening in the policy menu significantly increases the probability of the target rate actually increasing.

4.3 Inference

A key difficulty and econometric challenge for this paper is that of statistical inference: estimating the standard errors for the bounds derived in Section 3 is far from trivial. The key issue is that the estimators of the bounds are computed over non-sequential subsets of the original sample. For example, the series of variables could have originally been assumed to be sampled from an ARIMA process or a VAR, but now observations are non-sequentially selected and thus this new subset can no longer be modelled in the same fashion. Traditional techniques such as a simple bootstrap for independent observations can no longer be used to compute standard errors. Therefore, for the purpose of this paper, inference was not computed and is left as a potential addition for future work.

5 Dynamic Monetary Policy Effects: Empirical Results

I estimate the policy effects on three key macroeconomic variables: inflation, measured by the change in (100 times) the log of the Personal Consumption Expenditures Price Index (PCEPI); economic output, measured by the change in (100 times) the log of the Industrial Production (IP) index; and the unemployment rate. Causal responses to the policy treatment refer to the percentage change²⁰ in these outcome variables measured and accumulated from the

²⁰ Or percentage point change for unemployment.

month of the policy intervention to the respective horizon, where I assign both lower and upper bounds as derived in Section 3.

Our analysis delineates bounds on the impact of either increasing or decreasing the target funds rate by any amount in those respective categories (+0.50bp, +0.25bp, -0.25bp, -0.50bp). As in Angrist et al (2018), I extended this analysis out to a horizon of 24 months, and all responses are measured as cumulative percent changes. Furthermore, I do this for both symmetric and asymmetric bounds on the outcome variable: the latter corresponds to the bounds estimator of interest formulated in Section 3, while the former is merely a simplification where the outcome variable $Y_t(d)$ is now bounded by the symmetric closed interval $[K, K]$, imposing an extra restriction on the bounds²¹. By definition, as it can be shown trivially, this symmetric bound is less informative and should result in wider intervals – results are shown in Appendix A.3. Figure 3 plots these estimated responses, constructed using the third specification of the propensity score model. Table 3 formalizes and displays the results for each of these outcome variables at the selected 6, 12, 18 and 24 month horizons.

As is commonly accepted across macroeconomic theory, a tightening (easing) monetary policy is assumed to slow down the economy, and thus to yield a decrease (increase) in inflationary pressure, a decrease (increase) in economic output, and an increase (decrease) in the unemployment rate. Also, since the work of this paper yields bounds and not point estimates for dynamic monetary policy causal effects, I interpret whether these estimates are generally in line theoretical intuition by analyzing the proportion of the bounds that are either below or above zero, and the extent to which this occurs.

²¹ Now, the coefficient of the width of the bounds reduces to $4K > 2K_2 - 2K_1$. I display these results solely in the appendix A.3 for ease of exposition (Figure A.3.1 and Table A.3.3). The results are as expected: bounds are less informative, and thus intervals are larger in magnitude.

The first row of Figure 3 illustrates our causal effect bounds for the percentage change in inflation – these results can be seen in a clearer fashion in Table 3. An increase in the federal funds rate reduces the price level (decreases inflation) from the relevant counterfactual, and this effect is more significant in the short run – bounds are almost entirely contained below zero up to the 15-month horizon, whereas this is minimized for longer lags. This is a common finding in econometric literature on monetary policy effects, labelled as the “price puzzle”, where VAR-based studies (Sims 1992) indicate a short-run inflation increase with a tightening policy. Yet, two years out, an initial increase by the target funds rate is estimated to have an effect of prices somewhere in between a decrease of almost 0.4 percent and an increase of less than 0.2 percent. Conversely, the estimates for an easing monetary policy treatment are more inconclusive. Bounds are somewhat consistently symmetric around zero at all horizons, and the interval after 24 months for the effect on prices is $[-0.374, 0.346]$ percent. To put these results in perspective, it is worth noting that throughout our sample inflation is consistently low, hovering over the 2 percent annual rate.

Changing focus to the effect of monetary policy on output, we see much stronger and clearer effects. For a decrease in the target rate, the estimated bounds for IP are consistently above zero as expected by theory; and after two years the effect of decreasing the rate relative to the benchmark lies between a 0.4 and a 2.4 percent increase. Results for tightening policy are less evident but still interesting: there is an initial increase in production for the first year post-intervention, with the bound being almost strictly within the positive reals, whereas a steep decline follows in the subsequent 12-month period: at the 24-month horizon, we have bounds that are strictly negative, with an effect of between -1.9 and -0.1 percent in IP. This corresponds to about 0.04-0.80% annual growth rates in economic output.

Unemployment results also follow economic intuition. For both a target rate increase and decrease, bounds are respectively strictly above and below zero for the first 12-months. The causal response estimates for easing continue on this fashion up to the 24-month horizon, with final estimates falling between 0.4 and 1 percent declines in the unemployment rate. For tightening policy, the lower bound eventually falls below zero, but our estimates at the final horizon still fall within expectation, with most of the interval within the positive subset of the real line.

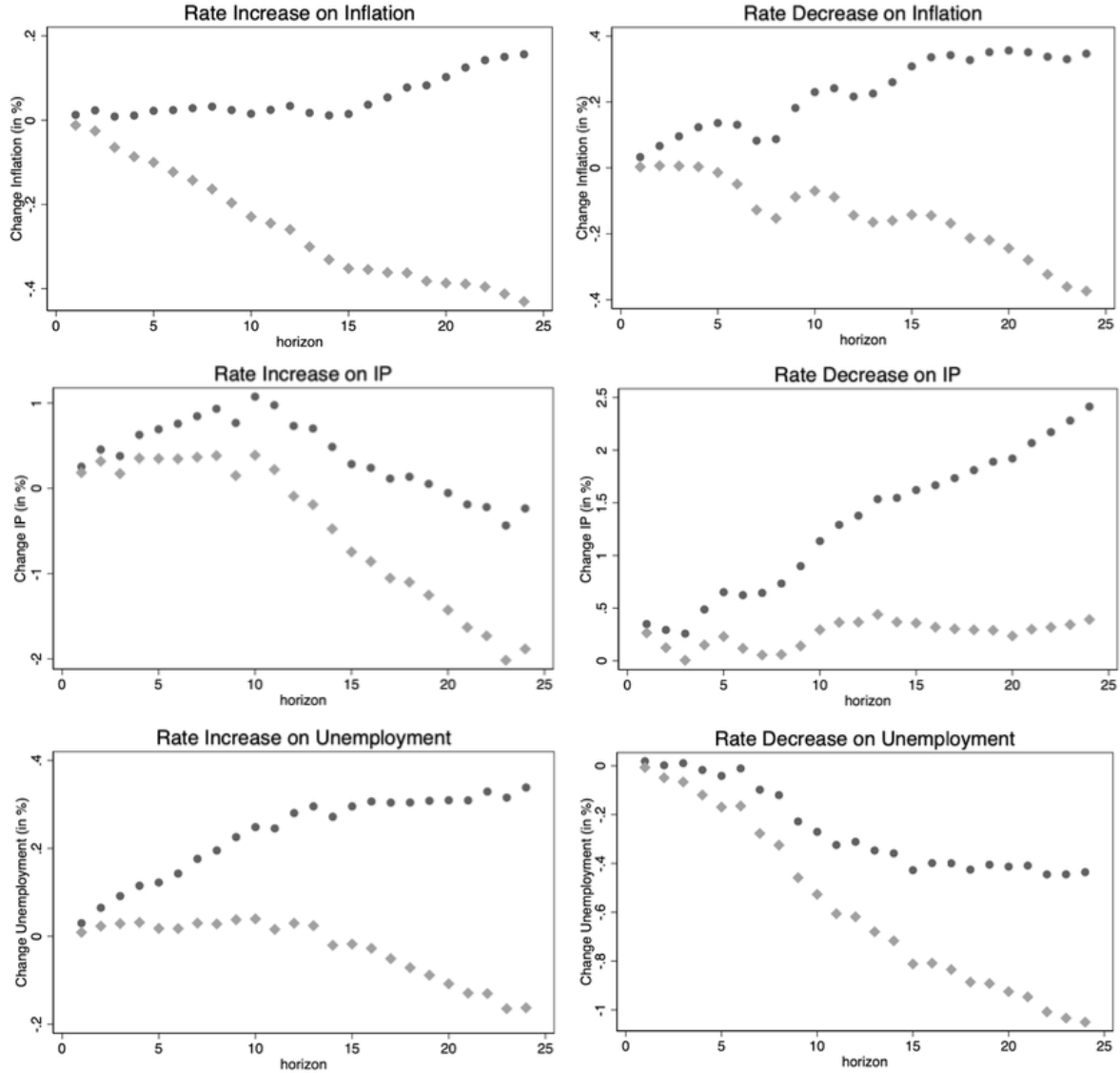
Our results, for the most part, fall in line with predictions from macroeconomic theory. Other studies have accounted for these asymmetries in the responses of variables to decreases or increases of the target rate, which we observe primarily with inflation (e.g. Hamilton and Jorda 2002, Angrist and Kuersteiner 2011). Also, the imposition of the extra restriction of symmetric outcome bounds in the derivation significantly increases the width of the causal effect estimates, as can be seen in Appendix A.3 (Table A.3.3 and Figure A.3.1).

**Table 3: Estimates of cumulated impulse responses at horizons 6, 12, 18 and 24 months
(Asymmetric Outcome Bounds)**

Horizon	Industrial Production		Unemployment		Inflation	
	Increase	Decrease	Increase	Decrease	Increase	Decrease
6	[0.345, 0.757]	[0.118, 0.623]	[0.017, 0.143]	[-0.164, -0.010]	[-0.123, 0.023]	[-0.049, 0.131]
12	[-0.092, 0.731]	[0.367, 1.377]	[0.029, 0.284]	[-0.619, -0.311]	[-0.259, 0.034]	[-0.144, 0.217]
18	[-1.099, 0.136]	[0.294, 1.809]	[-0.071, 0.384]	[-0.886, -0.425]	[-0.362, 0.078]	[-0.213, 0.327]
24	[-1.883, -0.235]	[0.392, 2.413]	[-0.162, 0.338]	[-1.050, -0.436]	[-0.432, 0.156]	[-0.374, 0.346]

Note: Values reported are cumulative changes in percent level of inflation and IP, as well as level rate changes in the unemployment rate. Interval displays both lower and upper bounds.

Figure 3: Estimated cumulative effects of target rate changes on macroeconomic variables.
(Asymmetric Outcome Bounds)



6 Summary and Conclusions

In line with some of the recent literature on the effect of monetary policy interventions, this study is able to semi-parametrically identify bounds on these dynamic causal effects by relaxing an important assumption of overlap. Motivation for the presence of no overlap in empirical research surrounding monetary policy effects is vast, and addressing it is a key

component in terms of statistical inference under this semi-parametric potential outcomes approach.

While still relying on selection-on-observables – that is, policy changes are independent of the potential outcomes conditional on observed covariates – which is a strong assumption but a necessary starting point (Angrist et al 2018), the results increase the generality of this recent semi-parametric literature by using the rich textual repository of the Federal Reserve to systematically violate overlap. The resulting bounds, which make use of the propensity score weighting (IPW) estimator, are able to also avoid the restriction of linearity which is commonplace in other parts of the literature (e.g. Sims 1980). Also, under this framework, we avoid the need of explicitly defining and constructing a generic series of monetary policy shocks, and our model implicitly defines these shocks as a purely identification question.

A severe setback of the study was that of inference: the nature of the problem makes this a particular daunting challenge. However, the identifiable and general estimates of bounds presented are in line with both economic theory and former empirical research. This paper should, hopefully, motivate more studies in macroeconometrics to adopt textual data as a potential aide in their research.

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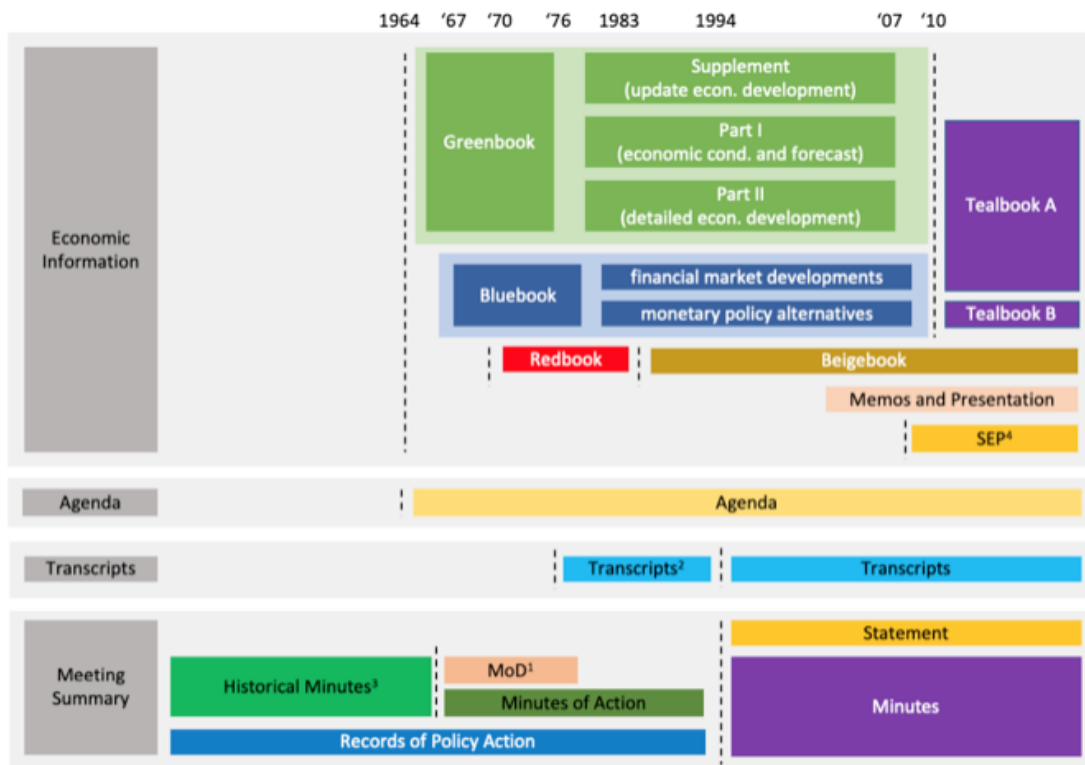
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A Appendix

A.1 Data Collection

Figure A.1.1: Evolution of Documents



¹Memoranda of Discussion

²Transcripts were prepared past 1994 and are lightly edited.

³Historical Minutes were prepared and released with the change in communication in 1967.

⁴Summary of Individual Economic Projections

Figure A.1.2: Time Series of Policy Alternatives and Fed Funds Target

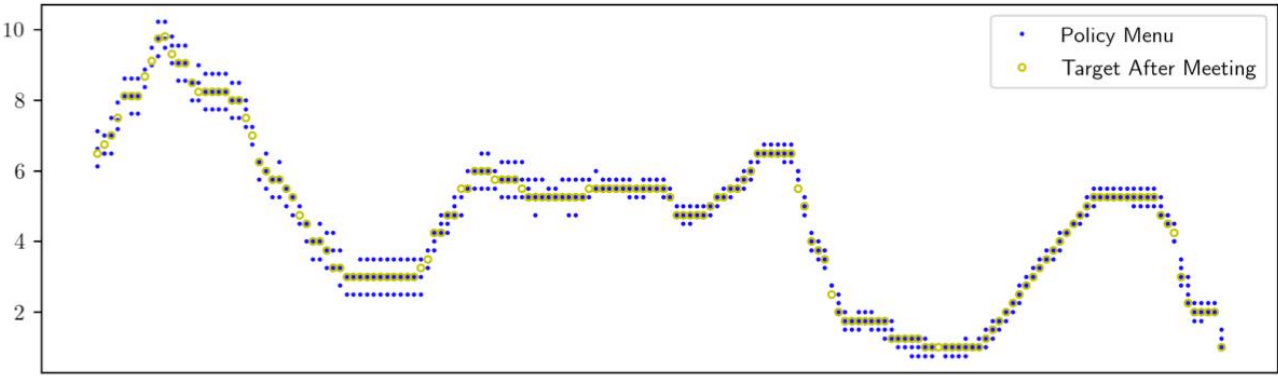


Figure A.1.3: Discrete Representation of Figure A.1.2

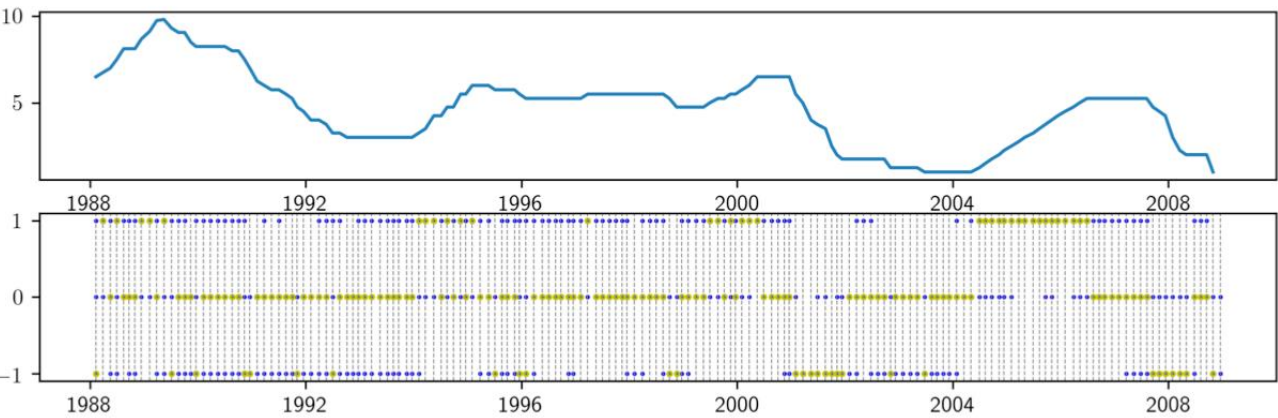


Table A.1.1: Policy Menu Breakdown into 25bps and their frequencies

Policy Options	Count	-0.5	-0.25	0.0	0.25	0.5	0.75
-0.25,-0.5	1	1	0	0	0	0	0
-0.25,-0.5,-0.75	2	2	0	0	0	0	0
-0.25,-0.5,0	9	2	4	3	0	0	0
-0.25,0	12	1	2	9	0	0	0
-0.25,0,0.25	11	0	1	10	0	0	0
-0.5,0	12	0	3	9	0	0	0
-0.5,0,0.25	2	0	2	0	0	0	0
-0.5,0,0.50	34	1	2	30	1	0	0
0	1	0	0	1	0	0	0
0,0.25	24	0	0	17	7	0	0
0,0.25,0.50	10	0	0	2	6	2	0
0,0.50	9	0	0	5	2	1	1
0.25	2	0	0	0	2	0	0
0.25,0.50	5	0	0	0	4	1	0

Table A.1.2: Summary Statistics for Target Changes (1989-2006)

Target Change	Count
-0.50	10
-0.25	20
0	77
0.25	22
0.5	5
Total	134

A.2 Derivation: Perfect correlation does not affect estimation

Note that:

$$E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x] = \int_{z \in \mathbb{Z}} E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x, Z_t = z] P(Z_t = z|X_t = x) dz$$

If X_t contains all the information about policy decision from Z_t , then we have that:

$$E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x, Z_t = z] = E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x]$$

And thus:

$$\begin{aligned} & \int_{z \in \mathbb{Z}} E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x, Z_t = z] P(Z_t = z|X_t = x) dz = \\ & E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x] \cdot \int_{z \in \mathbb{Z}} P(Z_t = z|X_t = x) dz = E[Y_{t+h}(d) - Y_{t+h}(0)|X_t = x] \end{aligned}$$

And so we are back to where we started. Thus, whether Z contains additional information about the policy choice D_t can be ignored in this derivation, as the estimation procedure will yield the same result even if we superfluously choose to incorporate Z into the covariate space.

A.3 Supplementary Results: Tables and Figures

Table A.3.1: Summary Statistics on Propensity Score Probabilities

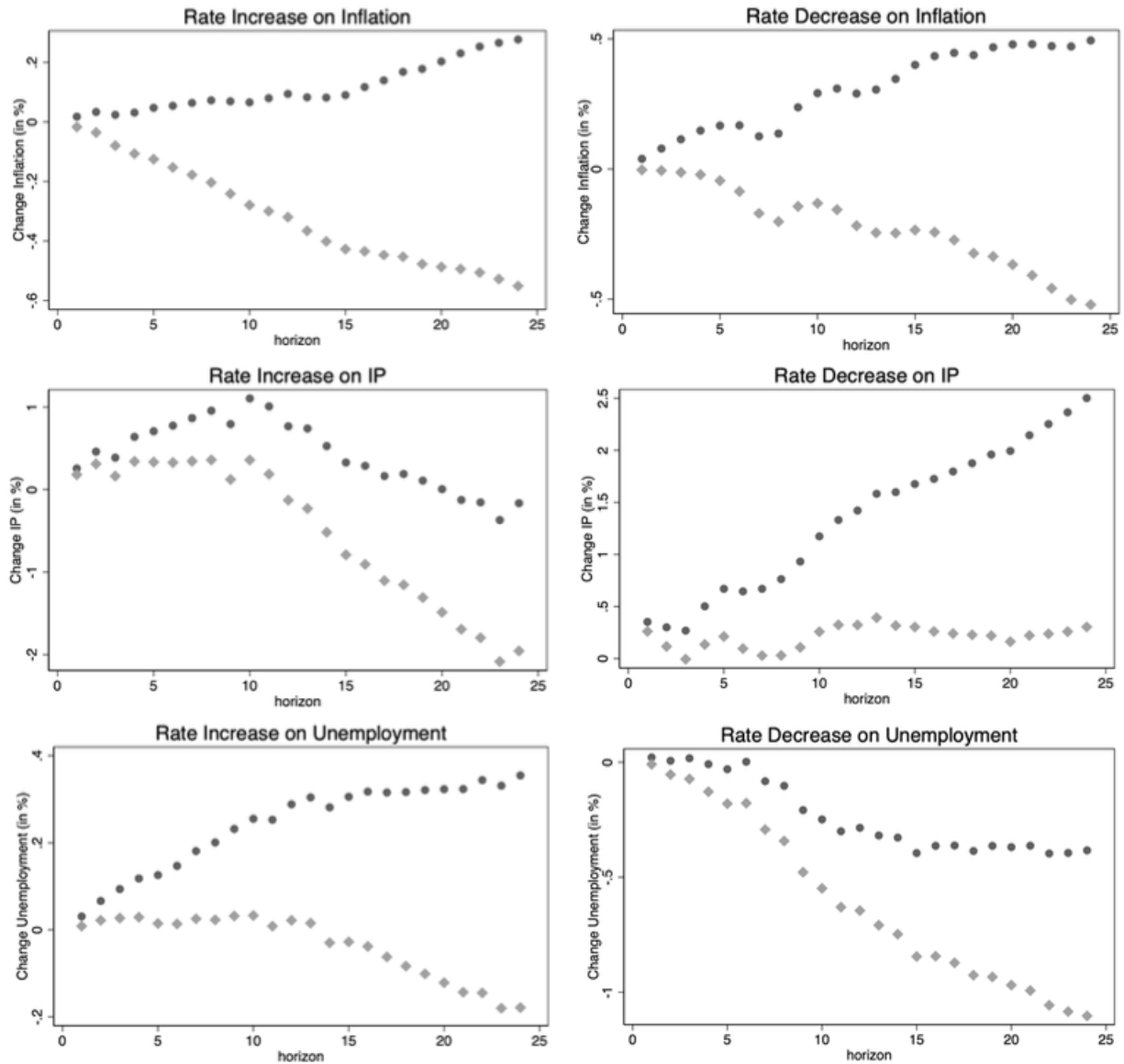
Variable	Observations	Mean	Std.Dev.	Min	Max
$P(D_t = 1 X_t = x)$	134	0.143	0.270	0.000	0.995
$P(D_t = 0 X_t = x)$	134	0.667	0.328	0.000	0.972
$P(D_t = -1 X_t = x)$	134	0.189	0.298	0.000	1.000

Table A.3.2: Ordered Probit Specifications (Marginal Effects on Decrease)

	(1)	(2)	(3)
Maintain in Menu			0.031 (0.055)
Ease in Menu			0.115** (0.051)
Tigthen in Menu			-0.181*** (0.042)
FFF		-0.966*** (0.125)	-0.755*** (0.131)
Inflation, Lag 1	-0.131 (0.120)	-0.146 (0.099)	-0.123 (0.098)
Inflation, Lag 2	-0.201* (0.124)	0.017 (0.107)	0.034 (0.106)
Unemp. rate, Lag 1	0.390*** (0.135)	0.229** (0.111)	0.215** (0.108)
Unemp. rate, Lag 2	0.557*** (0.133)	0.264** (0.113)	0.196* (0.112)
IP, Lag 1	-0.122*** (0.040)	-0.076** (0.032)	-0.041 (0.034)
IP, Lag 2	0.115*** (0.040)	0.069** (0.032)	0.035 (0.034)
Log Likelihood	-127.43	-88.28	-72.53
Observations	134	134	134

Note: Table reports marginal effects of listed covariates on the probability of an decrease of the target rate.
 ***/** corresponds to significance at the 90/95/99% confidence level.

Figure A.3.1: Estimated cumulative effects of target rate changes on macroeconomic variables.
(Symmetric Outcome Bounds)



Note: Estimates use data from July 1989 through January 2006. Upper and lower bounds from derivation reported using symmetric bounds on outcomes

Table A.3.3: Estimates of cumulated impulse responses at horizons 6, 12, 18 and 24 months
(Symmetric Outcome Bounds)

Horizon	Industrial Production		Unemployment		Inflation	
	Increase	Decrease	Increase	Decrease	Increase	Decrease
6	[0.328, 0.776]	[0.096, 0.645]	[0.013, 0.147]	[-0.177, 0.002]	[-0.153, 0.054]	[-0.086, 0.168]
12	[-0.128, 0.786]	[0.323, 1.421]	[0.022, 0.288]	[-0.645, -0.285]	[-0.319, 0.094]	[-0.218, 0.298]
18	[-1.153, 0.188]	[0.228, 1.875]	[-0.083, 0.316]	[-0.926, -0.385]	[-0.452, 0.168]	[-0.323, 0.43]
24	[-1.954, -0.164]	[0.384, 2.502]	[-0.178, 0.354]	[-1.103, -0.383]	[-0.551, 0.277]	[-0.521, 0.494]

Note: Values reported are cumulative changes in percent level of inflation and IP, as well as level rate changes in the unemployment rate. Interval displays both lower and upper bounds.